

Project Report: Hostel Management System

Name: Ilyan Khan

Roll No: 24CSE24

Department: Computer System Engineering

Introduction

This report outlines a simple, console-based Hostel Management System developed in C++. The project's primary objective is to provide a basic tool for hostel administrators to manage student records and room allocations efficiently. The system is designed using fundamental Object-Oriented Programming (OOP) principles to model real-world entities.

Core Features

The application provides the following functionalities through a text-based menu:

- **Student Registration:** Allows the user to add a new student by entering their name, ID, and contact number.
- **Student Record Updates:** Enables the modification of an existing student's details by searching for their ID.
- **Display All Students:** Lists the details of all registered students in the system.
- **Search:** Provides a facility to search for a specific student using their unique ID.
- **Room Allocation:** Allows the user to assign a room number to a registered student.

Technical Implementation

- **Language:** The project is implemented in C++.
- **Object-Oriented Design:** A `Student` class is used to encapsulate all the necessary data (**attributes**) and operations (**methods**) related to a student. This includes personal details as well as their assigned room number.
- **Data Structure:** A fixed-size array of `Student` objects is used as a simple in-memory database to store and manage student records.
- **User Interface:** The program uses a `while` loop and a `switch` statement to create a persistent and interactive command-line menu for the user.

Conclusion

The Hostel Management System successfully meets its objective of creating a functional, entry-level management tool. The project demonstrates a clear understanding of C++ basics and object-oriented concepts. While simple, it provides a solid foundation that can be expanded in the future with more advanced features like fee management, detailed room tracking, and data persistence by saving records to a file.